

7474 Decade Counter

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I. ABSTRACT

This paper shows how to use the 7474 D-Flip Flop ICs in a sequential circuit to realize a decade counter using arduino uno.

II. COMPONENTS

The required components list is given in Table: I., seven segment display is shown in Fig.1, IC 7447 diagram is shown in Fig.2 and 7474 D-Flip Flop pin diagram is shown in Fig-3.

Components	Value	Quantity
IC	7447	1
IC	7474	2
seven segment display		1
Arduino	UNO	1
Jumper Wires		50
Breadboard		1

TABLE I

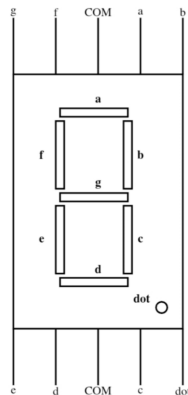


Fig. 1.

III. PROCEDURE

- 1) Make the connections of arduino, 7447 and two 7474 ICs according to Fig-4.
- 2) Block diagram of Decade Counter.
- 3) Execute the arduino code without any errors.
- 4) After upload the code into hardware setup using arduino IDE platform with hex file.

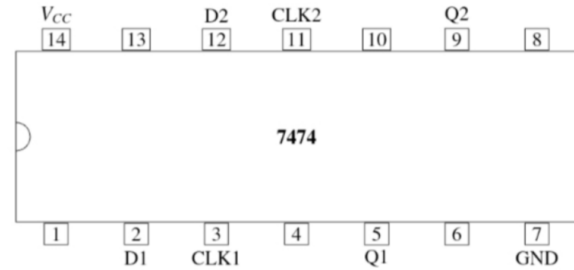


Fig. 2.



Fig. 3.

	INPUT					OUTPUT				CLOCK	5V				
	W	X	Y	Z	A	B	C	D							
Arduino	D6	D7	D8	D9	D2	D3	D4	D5	D13						
7474	5	9			2	12			CLK1	CLK2	1	4	10	13	
7474			5	9			2	12	CLK1	CLK2	1	4	10	13	
7447					7	1	2	6							16

Fig. 4.

IV. RESULTS

- 1) Download the code given in the link below and execute them to see the output as shown in Fig.6.
- 2) <https://github.com/madhureddy727/FWC-Assignments/tree/main/Assignment>

V. CONCLUSION

Hence implementation of 7474 IC Decade Counter on Seven segment display using arduino UNO is done.

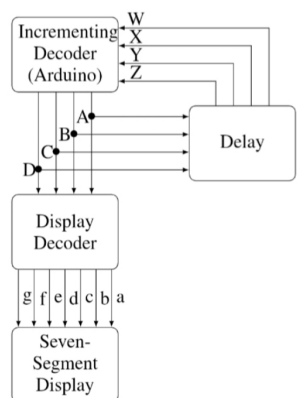


Fig. 5.

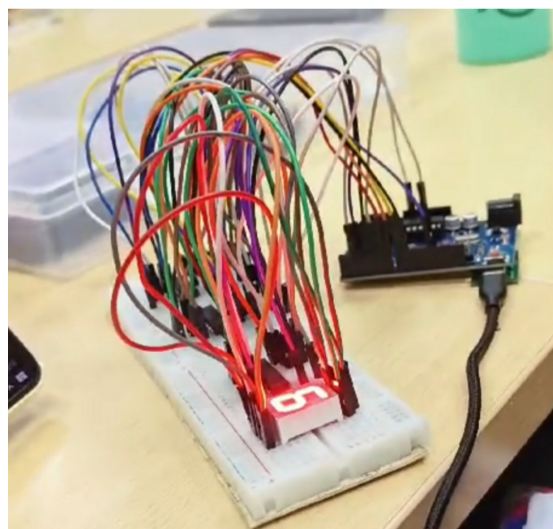


Fig. 6.